Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18CSDE721]

SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER-2021

ARTIFICIAL INTELLIGENCE

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1.	**a.**	What Do you mean by Intelligent Agents and Rational Agents in Artificial Intelligence? Elaborate with suitable Diagrams.	**07**
	b.	Differentiate between Toy problem and Real World problem in the context of AI. Explain with examples for each.	**07**
2.		Define the following terms:- agent, agent function, agent program, rationality, autonomy, reflex agent, model-based agent, goal-based agent, utility-based agent, learning agent.	**14**
3.	**a.**	What is Uninformed Search? What are different strategies under this search technique? Describe any two?	**07**
	b.	Elucidate the Informed (Heuristic) search strategies. Elaborate more on Greedy best-first search and Memory-bounded heuristic search.	**07**
		OR	
4.	**a.**	Describe minimax algorithms in detail.	**07**
	b.	What are Stochastic Games with respect to AI and give the detailed picture of any one game?	**07**
5.	**a.**	What are the advantages of Planning Graphs?	**07**
	b.	Define Classical Planning. Give one specific example	**07**
		OR	
6.	**a.**	Explain Hierarchical Planning in detail with respect to AI.	**07**
	b.	What is Multi-Agent Planning? Extend this explanation with suitable example	**07**
7.	**a.**	Define the terms: objects, predicates and variables in PROLOG	**07**
	b.	Debate on Rules and Unification in PROLOG.	**07**
		OR	
8.	**a.**	Describe how Best-first search technique can be applied to solve eight puzzle problems?	**07**
	b.	Explain the General principles of good programming.	**07**